

Explain how scientific knowledge is validated and refined, including the role of publication and peer review

Learning intention

- explain how scientific knowledge is validated and refined, including the role of publication and peer review.

Teach and model

- examining the scientific consensus supporting global warming.
- examining how Marie and Pierre Curie's discovery of new elements was validated.
- researching how JJ Thomson's discovery of the electron, Robert Millikan's oil drop experiment.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.